Tewodros Wondifraw Ayalew

Email: tewodrosayalew@uchicago.edu
Links: tedywond.github.io — github

Summary

I'm a PhD student in Computer Science at the University of Chicago, advised by Matthew Walter at the Robot Intelligence through Perception Laboratory (RIPL) at the Toyota Technological Institute at Chicago (TTIC).
I'm interested in how robots can learn about the physical world from video. My research focuses on world models, learning latent actions and dynamics from unlabeled videos, and learning reward signals from visual observations, with the goal of agents that learn from diverse real-world experience and transfer across tasks and embodiments.

Education

Sept 2022–Present — **PhD in Computer Science** at **The University of Chicago**, Chicago, USA
Advisor: Dr. Matthew Walter.
Focus: robot learning, reinforcement learning, world modeling, embodied intelligence.
robot-learning · reinforcement-learning · world-models

Sept 2018–Nov 2020 — **M.S. in Computer Science** at **The University of Saskatchewan**, Saskatoon, Canada
Advisor: Dr. Ian Stavness.
Thesis: *Unsupervised Domain Adaptation for Object Counting*.

Sept 2011–July 2016 — **BSc in Software Engineering** at **Addis Ababa University**, Addis Ababa, Ethiopia

Research Experience

Oct 2021–Sept 2022 — **Machine Learning Engineer** at **Global Institute for Food Security**, Saskatoon, Canada
Mentors: Dr. Ian Stavness and Professor Leon Kochian.
Developed computer vision approaches to phenotype plant roots from field imagery, targeting yield-predictive feature extraction.
computer-vision · phenotyping · deep-learning

Jan 2021–Sept 2021 — **Machine Learning Engineer** at **Honda R&D / University of Saskatchewan**
Mentors: Dr. Ian Stavness and Dr. Masaki Yamazaki.
Designed and evaluated unsupervised domain adaptation approaches for robust semantic segmentation under adverse weather conditions, supporting autonomous robot navigation.
domain-adaptation · semantic-segmentation · autonomous-driving

Publications

*Equal contribution. First-author papers in bold author name.

2026 **CLAW: Learning Continuous Latent Action World Models via Adversarial Latent Regularization**
Tewodros W. Ayalew, Matthew Jeung*, Samuel Wheeler*, Xiao Zhang, Andre de la Cruz Arce, Kaylene Stocking, Michael Maire, Matthew R. Walter
CoRL 2026 Workshop PhysWM
WM@Booth, 2026 **[Lightning Talk]**
Midwest Robotics Workshop (MWRW), 2026 **[Contributed Talk]**
Project · arXiv · Video

2025 **PROGRESSOR: A Perceptually Guided Reward Estimator with Self-Supervised Online Refinement**
Tewodros W. Ayalew, Xiao Zhang*, Kevin Yuanbo Wu*, Tianchong Jiang, Michael Maire, Matthew Walter
ICCV, 2025
Project · Code

2026 **What Are We Actually Benchmarking in Robot Manipulation?**
Tianchong Jiang, Xiangshan Tan*, Samuel Wheeler*, Luzhe Sun*, **Tewodros W. Ayalew***, Matthew Walter
CoRL, 2026 **[Oral] [Best Paper Award Finalist]**
IROS 2026 RGMCW Workshop **[Best Abstract Award]**
Project · arXiv · Code · Video

2025 **Enabling End Users to Program Robots Using Reinforcement Learning**
Tewodros W. Ayalew, Jennifer Wang, Michael L. Littman, Blase Ur, Sarah Sebo
HRI, 2025

2021 **Automatic Microplot Localization Using UAV images and a Hierarchical Image-based Optimization Method**
Sara Mardanisamani, **Tewodros W. Ayalew**, Minhajul Arfin Badhon, Nazifa Azam Khan, Gazi Hasnat, Hema Duddu, Steve Shirtliffe, Sally Vail, Ian Stavness, Mark Eramian
Plant Phenomics, 2021

2020 **Unsupervised Domain Adaptation for Plant Organ Counting**
Tewodros W. Ayalew, Jordan Ubbens, Ian Stavness
ECCV Workshop on Computer Vision Problems in Plant Phenotyping (CVPPP), 2020 **[Best Paper Award]**

2020 **AutoCount: Unsupervised Segmentation and Counting of Organs in Field Images**
Jordan Ubbens, **Tewodros W. Ayalew**, Steve Shirtliffe, Anique Josuttes, Curtis Pozniak, Ian Stavness
ECCV Workshop on Computer Vision Problems in Plant Phenotyping (CVPPP), 2020

Selected Talks

2026 **CLAW: Learning Continuous Latent Action World Models via Adversarial Latent Regularization**
WM@Booth 2026 — Lightning talk

2026 **CLAW: Learning Continuous Latent Action World Models via Adversarial Latent Regularization**
Midwest Robotics Workshop (MWRW) 2026 — Contributed talk

2025 **Enabling End Users to Program Robots Using Reinforcement Learning**
International Conference on Human-Robot Interaction (HRI 2025)

2020 **Unsupervised Domain Adaptation for Plant Organ Counting**
ECCV Workshop on Computer Vision Problems in Plant Phenotyping (CVPPP 2020)

Honors & Scholarships

2026 **Best Paper Award Finalist**
"What Are We Actually Benchmarking in Robot Manipulation?," CoRL 2026

2026 **Best Abstract Award**
IROS 2026 Workshop on Reproducible Benchmarking of Robotic Grasping and Manipulation

2025 **CIFAR Inclusive AI Scholarship**
CIFAR Deep Learning and Reinforcement Learning Summer School

2022 **Crerar Fellowship**
Computer Science PhD Fellowship, The University of Chicago

2020 **Best Paper Award**
"Unsupervised Domain Adaptation for Plant Organ Counting," ECCV Workshop (CVPPP)

2020 **Best Poster Award**
5th Annual Plant Phenotyping and Imaging Research Center Symposium

2019 **Best Poster Award**
4th Annual Plant Phenotyping and Imaging Research Center Symposium

2018 **Graduate Teaching Fellowship**
The University of Saskatchewan

Volunteer Work

Summer 2018 **Teaching Programming for High School Students**
Designed and delivered a summer course introducing Scratch and Python to high school students.